Hannah Lee

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Education

University of Washington *Sept 2023 – June 2024*
M.S. Computer Science and Engineering GPA: 3.97/4.0

- **Advisor:** Oren Etzioni
- **Thesis:** Identifying Modern Deepfakes: Bringing Fake Image Detection into the Wild

University of Washington *Sept 2019 – June 2023*
B.S. Computer Science; Applied Mathematics GPA: 3.98/4.0

- **Advisor:** Shwetak Patel; **Mentored by:** Jason Hoffman
- **Thesis:** Determining Input Image Quality for Smartphone Detection of Anemia
- *Magna cum laude, Interdisciplinary Honors Program*

Experience

Software Engineer *Redmond, WA*
Microsoft *Sept 2024 – Present*

- Responsible for designing and implementing key software tools to drive down network repair times in Microsoft's datacenters, utilizing Azure Kubernetes and microservice design patterns.
- Designed and implemented a new onboarding procedure to streamline critical environment spare inventory for new datacenters.

Machine Learning Intern *Seattle, WA*
TrueMedia.org *June 2024 – Aug 2024*

- Developed and trained a novel classification model that distinguishes between AI manipulated or otherwise digitally altered images from real images with over 97% accuracy on academic datasets and over 80% on unseen, in-the-wild examples, contributing to TrueMedia's efforts to fight political misinformation.
- Consistently conducted literature reviews and analyses of open-source systems to determine the state-of-the-art in deepfake detection, working with a small research team.

Software Engineering Intern *Redmond, WA*
Microsoft *June 2023 – Sept 2023*

- Designed and implemented Azure Function Jobs and UI tools to automate several manual processes for datacenter technicians, supporting the rapid expansion of Microsoft's cloud investments.

Undergraduate Researcher *Seattle, WA*
UbiComp Lab, University of Washington *Sept 2021 – June 2023*

- Researched and developed a classification model to identify high quality patient images to aid in a larger pipeline estimating Hemoglobin levels from fingernail images, using classical and modern machine learning techniques and a custom-built labeling tool to label hand images.
- Assisted in research development of a remote health sensing system to monitor Central Venous Pressure in heart failure patients using smartphone cameras and computer vision and signal processing techniques.

Software Engineering Intern *San Francisco, WA*
Salesforce *June 2022 – Sept 2022*

- Developed a Salesforce Lightning Web Component in Salesforce Commerce Cloud to display "Frequently Bought Together" products to encourage bundling of items and provide upselling opportunities.

Software Engineering Intern *Bellevue, WA*
T-Mobile US *June 2021 – Sept 2021*

- Built internal tools to reduce administrative task hours by 50% for T-Mobile Supply Chain engineers and automated testing for the team's IBM Operational Decision Manager (ODM) APIs.

Publications

Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024
Nuria Alina Chandra, Ryan Murtfeldt, Lin Qiu, Arnab Karmakar, Hannah Lee, Emmanuel Tanumihardja, Kevin Farhat, Ben Caffee, Sejin Paik, Changyeon Lee, Jongwook Choi, Aerin Kim, Oren Etzioni.
Preprint, May 2025.

MINT-1T: Scaling Open-Source Multimodal Data by 10x: A Multimodal Dataset with One Trillion Tokens
Anas Awadalla, Le Xue, Oscar Lo, Manli Shu, Hannah Lee, Etash Kumar Guha, Matt Jordan, Sheng Shen, Mohamed Awadalla, Silvio Savarese, Caiming Xiong, Ran Xu, Yejin Choi, Ludwig Schmidt.
NeurIPS 2024, *Datasets and Benchmarks Track*, Vancouver, Cananda, Dec. 2024.

The Tug-of-War Between Deepfake Generation and Detection
Hannah Lee, Changyeon Lee, Kevin Farhat, Lin Qiu, Steve Geluso, Aerin Kim, Oren Etzioni.
Data-centric Machine Learning Workshop at the 41^{st} International Conference on Machine Learning, Vienna, Austria, Jul. 2024.

Artificial intelligence-enabled non-invasive ubiquitous anemia screening: The HEMO-AI pilot study on pediatric population
Daniel Gordon, Jason Hoffman, Keren Gamrasni, Yotam Barlev, Alex Levine, Tamar Landau, Ronen Shpiegel, Avishai Lahad, Ariel Koren, Carina Levin, Osnat Naor, Hannah Lee, Xin Liu, Shwetak Patel, Gilad Chayen, Michael Brandwein.
DIGITAL HEALTH, 2024.

Recognition

Bob Bandes Memorial Award for Excellence in Teaching (Honorable Mention) *Paul G. Allen School, University of Washington* — *June 2024*

Boeing Excellence Award *Department of Applied Mathematics, University of Washington*. Awarded to the top performing graduate. — *June 2023*

Dean's List (Annual) *University of Washington* — *Sept 2019 - June 2023*

College Honors *Honors Program, University of Washington* — *June 2023*

Presentations

The Tug-of-War Between Deepfake Generation and Detection (Poster)
Data-Centric Machine Learning Research Workshop at the International Conference on Machine Learning, Vienna, Austria, July 2024.

Determining Input Image Quality for Smartphone Detection of Anemia
26th Annual Undergraduate Research Symposium, University of Washington, Seattle, WA, May 2023.

Monitoring Central Venous Pressure Using Smartphone Video
25th Annual Undergraduate Research Symposium, University of Washington, Seattle, WA, May 2022.

Teaching Experience

Graduate Teaching Assistant *University of Washington*

- Responsible for leading weekly staff meetings and delegating tasks; preparing course assignments, materials, and exams; maintaining and improving course code infrastructure; holding office hours; and monitoring online discussion boards.
- Worked with Ludwig Schmidt, Jamie Morgenstern, Kevin Jamieson, Simon Du, Hunter Schafer, and Matthew Golub over six quarters:
  - *Spring 2024* CSE 599N Machine Learning for Neuroscience
  - *Winter 2024* CSE 446/546 Machine Learning (Head Teaching Assistant)
  - *Autumn 2024* CSE 446/546 Machine Learning (Head Teaching Assistant)

Undergraduate Teaching Assistant *University of Washington*

- *Spring 2023* CSE 446/546 Machine Learning (Head Teaching Assistant)
- *Winter 2023* CSE 446/546 Machine Learning (Co-Head Teaching Assistant)

- *Autumn 2022* CSE 446/546 Machine Learning

Skills

Languages: Python, C#, Java, TypeScript, C++

Technologies: PyTorch, OpenCV, NumPy, SciPy, .NET, Microsoft SQL Server, Docker, Azure Kubernetes Service